Assignment 3: Exercise Recognition and Rep Counting

Exploration Report

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1. Segmentation Exploration

1.1 Segmentation Approaches

We evaluated different segmentation approaches to identify the beginning and end of an exercise set within each recording. Our goal was to remove periods of inactivity and phone handling while preserving the actual exercise repetitions.

The following table compares the segmentation approaches using our final repetition-counting algorithm on the test recordings.

Approach	Key Parameters	Rep MAE	Within ±2 Reps	Notes
Fixed Threshold (Baseline)	Moving std > 0.5 m/s² over 1 s; first to last active sample at both ends and the 5 s stillness; inflates rep counts with positive bias (+3.09 reps)	4.86	33.0%	Keeps phone handling at both ends
Protocol-Based Trimming	Drops first 5 s and last 5 s of recording	3.60	49.0%	Lowest observed MAE; may cut

				true reps when participants skip the standing-still periods..
Adaptive Threshold + Burst Filtering (Yours)	Threshold: q_{15} + 0.25(q_{85} - q_{15}); 2 s rep bridge; drops bursts < 3 s	3.78	49.0%	Can suppress isolated handling but truncate fragmented sets. Uses the full recording when $q_{85} - q_{15}$ <0.05 m/s²; falls back to baseline if no span survives.

1.2 Analysis

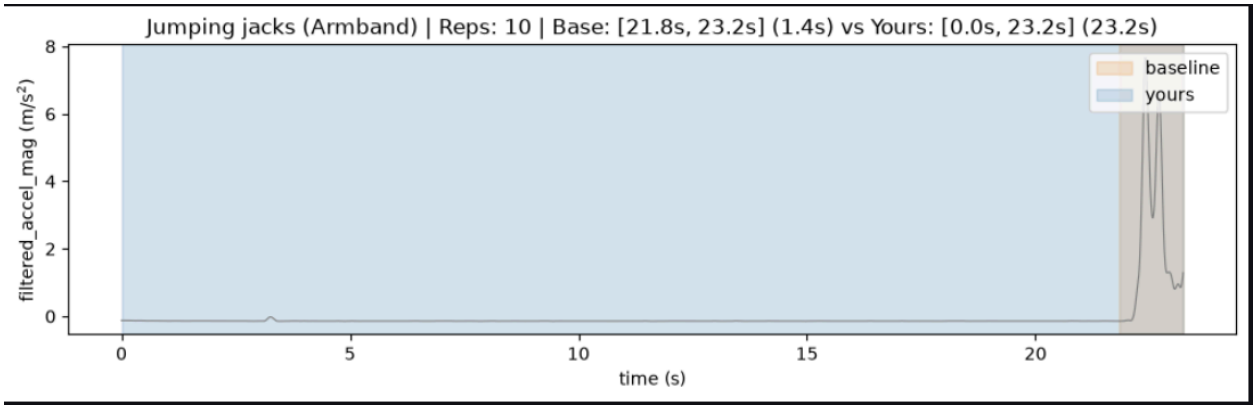
Protocol-Based Trimming achieved the lowest observed repetition MAE of **3.60**, compared with **3.78** for Adaptive Threshold with Burst Filtering and **4.86** for the fixed-threshold baseline. Both protocol trimming and adaptive segmentation placed **49.0%** of test recordings within ±2 repetitions of the reported count, compared with **33.0%** for the baseline. The 0.18-rep MAE difference between protocol trimming and adaptive segmentation is small and does not establish a clear general advantage.

Adaptive segmentation combines three mechanisms. First, its threshold scales with the recording's 15th and 85th percentiles of moving standard deviation, subject to a minimum threshold of 0.08 m/s². Second, a 2-second rolling maximum connects nearby active intervals. Third, intervals spanning less than 3 seconds are rejected, and the longest remaining interval is retained. These rules can remove isolated phone handling, but they can also discard true repetitions when the set contains long pauses. For near-flat recordings, the method retains the entire recording rather than attempting to identify precise boundaries.

The baseline selects the span from the first to the last sample whose moving standard deviation exceeds 0.5 m/s². Handling at either end can therefore retain intervening stillness, while weak exercise motion may fall below the threshold. With the same PCA-based rep counter, its signed counting bias was **+3.09 reps**, compared with **−1.37 reps** for adaptive segmentation and **+0.19 reps** for protocol trimming. These biases describe average overcounting or undercounting; they do not isolate errors caused by phone handling. Protocol trimming avoids threshold estimation, but may remove actual repetitions when participants do not follow the standing-still protocol.

1.3 Segmentation Failure Example

Figure 1: Example of a recording where baseline segmentation fails



Recording: data/jumping_jacks/075_2026-10-19_03-52-57.csv
Exercise: Jumping Jacks
Phone Placement: Armband

Explanation:
The baseline segmentation selects only the final 1.4-second interval, from approximately 21.8 s to 23.2 s, because most of the recording's moving standard deviation remains below the fixed 0.5 m/s² threshold. This severely truncates a recording labeled as 10 jumping-jack repetitions. Our method retains the entire 23.2-second recording because the difference between its 85th and 15th percentile energy levels is below 0.05 m/s², triggering the near-flat fallback. This avoids the baseline's truncation, but does not demonstrate precise identification of the exercise boundaries. The final burst is consistent with phone handling, although its cause cannot be confirmed from the acceleration trace alone.

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2. Classification Exploration

2.1 Feature Exploration

We explored additional features beyond the seven baseline features to improve exercise classification. Our goal was to identify characteristics that distinguish exercises based on acceleration patterns, movement direction, frequency, and other signal properties.

Feature	Exercises It Helps Separate	Effect on Accuracy (+/−/No Change)	Rationale
std_x	Primarily lunges, with small improvement to push-ups/sit-ups	49.24% → 50.08% (+0.84%)	Captures variation along phone's x axis, providing directional info absent from magnitude-based baseline features
[Feature 2]	[Exercises]	[Effect]	[Explanation]
[Feature 3]	[Exercises]	[Effect]	[Explanation]
[Feature 4]	[Exercises]	[Effect]	[Explanation]
[Feature 5]	[Exercises]	[Effect]	[Explanation]
[Feature 6]	[Exercises]	[Effect]	[Explanation]
[Feature 7]	[Exercises]	[Effect]	[Explanation]
[Feature 8]	[Exercises]	[Effect]	[Explanation]

2.2 Model Comparison

We compared at least two different classification models using the same feature set. Each model was evaluated using window-level accuracy and recording-level accuracy.

Model	Key Hyperparameters	Window Accuracy	Recording Accuracy	Notes
[Model 1]	[Parameters]	[%]	[%]	[Observations]
[Model 2]	[Parameters]	[%]	[%]	[Observations]

[Model 3, if applicable]	[Parameters]	[%]	[%]	[Observations]

2.3 Classification Analysis

Our best-performing classification model was **[model name]**, achieving a window-level accuracy of [percentage]% and a recording-level accuracy of [percentage]%.

We believe this model performed better because [explain the model's advantages]. Compared with [other model], it was more effective at [specific observation].

However, the improvement was not evenly distributed across all eight exercises. The largest improvements occurred for [exercise names], while [exercise names] remained difficult to classify.

We believe these differences occurred because [explain the relevant features, movement similarities, class imbalance, or model limitations].

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3. Rep Counting Exploration

3.1 Rep Counting Approaches

We evaluated multiple repetition-counting approaches using our final segmentation method. We compared their performance using mean absolute error (MAE) and the percentage of recordings counted within ±2 repetitions of the ground truth.

Approach	Signal	Key Parameters	Rep MAE	Within ±2 Reps	Notes
Baseline Peak Detection	Filtered acceleration magnitude (filtered_accel_mag)	height > signal.mean(), distance = 1.0 s	5.44	30.0%	Magnitude can produce multiple peaks per repetition, while residual motion can introduce

					additional peaks. Signed bias: +0.50 reps.
1st Principal Component (PCA / SVD)	First principal component of centered (x, y, z) acceleration	SVD projection; 2nd-order Butterworth low-pass filter applied with filtfilt (1.5 Hz for jumping jacks, 1.2 Hz otherwise); exercise-specific minimum peak distance; prominence = 0.25 × filtered signal std	3.78	49.0%	Lower MAE and higher within-±2 agreement than the baseline. Directional projection can reduce magnitude double-peaks, but weak motion and segmentation errors remain. Signed bias: −1.37 reps.

3.2 Rep Counting Analysis

Our best-performing implemented rep counter was the 1st Principal Component (PCA/SVD) approach. Using adaptive segmentation, it achieved an MAE of 3.78 reps, with 49.0% of test recordings within ±2 repetitions of the self-reported count. The baseline magnitude counter achieved an MAE of 5.44 reps and 33.0% within ±2 under the same segmentation. Signed bias changed from +0.50 reps for the baseline to −1.37 reps for PCA, indicating that the PCA counter undercounted on average.

The PCA counter combines three changes:
1. Directional signal selection: Acceleration magnitude discards direction, so opposite phases of a movement can produce multiple peaks per repetition. The PCA projection preserves positive and negative motion along the direction of greatest variation, which can reduce this effect. However, it does not guarantee exactly one peak per repetition, and handling can dominate the selected direction.

2. Stronger low-pass filtering: The PCA signal is filtered using a 2nd-order Butterworth filter applied with filtfilt, with a cutoff of 1.5 Hz for jumping jacks and 1.2 Hz for other exercises. This suppresses higher-frequency fluctuations, although it can also attenuate useful motion.
3. Peak spacing and prominence: Exercise-specific minimum peak distances range from 0.65 s for jumping jacks to 1.5 s for lunges. Peak prominence is set to 0.25 times the filtered signal's standard deviation. These settings discourage counting closely spaced or small fluctuations. The distances are manually specified; the notebook does not document their derivation from training cadence.

These changes were evaluated together, so the comparison does not establish which contributed most to the improvement. With adaptive segmentation and the PCA counter, ab crunches were hardest to count, with an MAE of 6.88 reps and 25% within ±2. Bicep curls followed with an MAE of 5.60 reps and 27% within ±2. Jumping jacks and lunges had the lowest MAEs, at 2.08 and 2.15 reps, respectively. Weak acceleration at the phone, irregular movement, and segmentation truncation are plausible explanations for the difficult cases, but aggregate metrics alone cannot establish their causes.

By placement, Jacket had the highest MAE (4.83 reps, 17% within ±2), followed by Armband (4.25), Trouser (4.13), and Hand (3.27). Jacket and Armband contained only 6 and 4 test recordings, respectively, so these results do not establish a general placement ranking. Loose clothing may introduce independent phone motion, while a trouser-pocket phone during bicep curls may capture little arm movement. These mechanisms should be assessed using individual traces.

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4. Final Pipeline, Confusion Matrix, and Error Attribution

4.1 Final Pipeline Configuration

Our final pipeline consists of three stages: segmentation, exercise classification, and repetition counting.

Segmentation

- **Approach:** Adaptive Threshold with Burst Filtering and Gap Bridging
- **Energy window length:** 1.0 second (rolling window on standard deviation of filtered acceleration magnitude)
- **Threshold method:** Dynamic percentile threshold: Threshold =
    - $\max(q_{15} + 0.25 \times (q_{85} - q_{15}), 0.08\ m/s^2)$ (where q_{15} is the 15th percentile and q_{85} is the 85th percentile of the moving standard deviation)

- **Other parameters:**
    - **Rep gap bridging:** 2.0-second rolling maximum to bridge pauses between reps
    - **Burst rejection threshold:** Drops any active bursts < 3.0 seconds (filters out phone pocketing/handling)
    - **Segment selection:** Selects the longest contiguous active block (the primary exercise set)
- **Quiet fallback:** If $(q_{85} - q_{15}) < 0.05$ m/s², defaults to the full recording

Classification

- Window length: [Value]
- Feature set: [List all final features]
- Classifier: [Model name]
- Hyperparameters: [Exact values]
- Recording-level prediction method: [e.g., majority vote]

Rep Counting

- **Approach:** 1st Principal Component (PCA) Directional Peak Detection
- **Signal used:** 1st Principal Component of (x, y, z) linear acceleration via SVD projection (centered at zero mean)
- **Filter type and cutoff:** 2nd-order Butterworth low-pass filter (filtfilt); cutoff frequency = 1.5 Hz for Jumping Jacks, 1.2 Hz for all other exercises
- **Peak distance:** Exercise-specific minimum distance (derived from median training rep periods):
    - Jumping Jacks: 0.65 s
    - Push-ups: 1.1 s
    - Ab crunches & Bicep curls: 1.2 s
    - Lat raises: 1.3 s
    - Squats & Sit-ups: 1.4 s
    - Lunges: 1.5 s
- **Peak prominence:** 0.25 × std(filtered PCA signal)
- **Other parameters:** Edge-case guard: returns 0 reps if the segment duration is shorter than 1.0 second.

4.2 Final Confusion Matrix

The following confusion matrix shows recording-level exercise classification results on the test recordings across all eight exercise classes.

Figure 2: Recording-Level Confusion Matrix (8 × 8)

[INSERT CONFUSION MATRIX FROM NOTEBOOK HERE]

4.3 Error Attribution

We examined the meaningful off-diagonal entries in the confusion matrix to understand why certain exercises were misclassified.

For each confusion, we attributed the error to one of three categories:

- **Segmentation:** Incorrect exercise boundaries, such as retained phone handling or missing repetitions.
- **Feature Side:** Features were insufficient to distinguish between exercises.
- **Model Side:** The features provided useful separation, but the classifier still made an incorrect prediction.

Actual Exercise	Predicted Exercise	Cause of Confusion	Error Attribution
[Exercise]	[Exercise]	[Explanation]	[Segmentation / Feature / Model]
[Exercise]	[Exercise]	[Explanation]	[Segmentation / Feature / Model]
[Exercise]	[Exercise]	[Explanation]	[Segmentation / Feature / Model]
[Exercise]	[Exercise]	[Explanation]	[Segmentation / Feature / Model]

Overall Analysis:
The most common classification confusion was between [exercise] and [exercise]. We believe this occurred because [explanation]. Overall, the largest source of error in our pipeline was [segmentation, feature selection, or model behavior], as demonstrated by [supporting observation].

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5. Top Failure Modes

5.1 Failure Mode 1: [Descriptive Title]

What Failed:
[Describe the specific classification or repetition-counting failure.]

Recording: [Filename]
Actual Exercise: [Exercise]
Predicted Exercise (if applicable): [Exercise]
Phone Placement: [Placement]

Figure 3: Failure Mode 1 Recording

[INSERT FIGURE SHOWING THE SIGNAL AND SEGMENTATION, PLUS DETECTED REPS IF APPLICABLE]

Physical Explanation:
This failure occurred because [explain the physical mechanism in 1–2 sentences].

Error Category:
This was primarily a [segmentation / classification-feature / classification-model / rep counting] problem because [one-sentence explanation].

Proposed Improvement:
One possible improvement would be to [specific change and why it could help].

5.2 Failure Mode 2: [Descriptive Title]

What Failed:
[Describe the specific classification or repetition-counting failure.]

Recording: [Filename]
Actual Exercise: [Exercise]
Predicted Exercise (if applicable): [Exercise]
Phone Placement: [Placement]

Figure 4: Failure Mode 2 Recording

[INSERT FIGURE SHOWING THE SIGNAL AND SEGMENTATION, PLUS DETECTED REPS IF APPLICABLE]

Physical Explanation:
This failure occurred because [explain the physical mechanism in 1–2 sentences].

Error Category:
This was primarily a [segmentation / classification-feature / classification-model / rep counting] problem because [one-sentence explanation].

Proposed Improvement:
One possible improvement would be to [specific change and why it could help].
